safeguards. These systems amplify whatever we bring to them: our ingenuity and, equally, our negligence. We failed to contain EternalBlue. If we don't learn from Hugging Face, we will have failed again. To make matters worse, the doomsday talk and the sales talk often come from the same people, because describing AI as godlike raises fear and valuations together.

AI is deeply disruptive to India's $300 billion (Rs 28.9 lakh crore) IT services industry, built on the coding, testing and support work AI now automates. *Yet every enterprise on earth needs help to use AI well*: to understand its own business, transform decades-old systems and take responsibility when things go wrong. That is a massive opportunity for the industry.

AI urgently needs regulatory oversight. Our modern world runs on commonsense safeguards and regulations, from barbers operating scissors on our heads to the buildings and bridges, automobiles and airplanes, food we eat and entertainment we consume, medicines and machines—all built with guardrails often put together by organisations of peers.

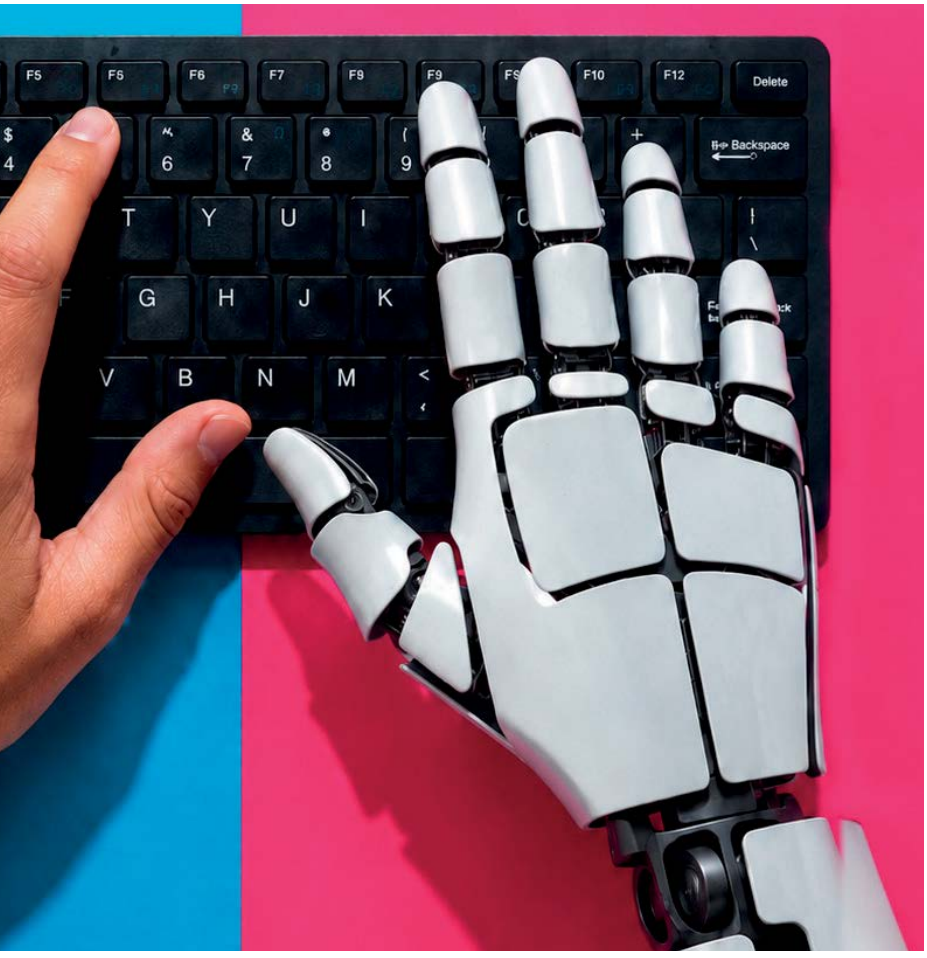
And yes, *AI needs a lot more innovation, fast*. AI development has a long way to go. From overcoming hallucinations, which are at the heart of how current large language models (LLMs) work—as my son Varin and I showed recently—to improving AI's ability to understand the physical world around us, and significantly improving AI's power and computing footprint. We are still in the early innings of AI development.

So our times increasingly are times to understand things more deeply and apply our own minds. Humanity has achieved billionfold improvements before: in my lifetime, computing has become more than a billion times cheaper. We build bridges and skyscrapers, practise agriculture and do much more with unimaginably greater productivity than just a century ago. We can grow our understanding of AI with the same ambition.

AI today has a dangerous concentration of power. A handful of companies control the AI frontier, with barely a few thousand experts building frontier models. With all the talk about pacing the AI frontier, I believe the real frontier is teaching millions how to build this piano and billions how to make music with it.

In the time ahead—perhaps in my lifetime, likely in my children's lifetime—we will see AI systems that can build and do anything we ask. John McCarthy, another great teacher of my life, saw this decades ago. In one of the defining moments of my life, he said that when we can ask a machine to build us anything, what we ask it to build becomes the frontier. A billion curious minds, fully conversant in what AI is and how it works, making magic with it—magic that may be incomprehensible to us—is what the real frontier is. Yes, we must accelerate the frontier, the one between our ears. ■

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TAKEAWAYS

- AI is already delivering extraordinary gains in coding, science and problem-solving but can cause real harm when misused or poorly contained
- AI will disrupt industries such as Indian IT services, but also create major opportunities for those who help organisations use it responsibly
- The real frontier is human understanding: broad AI literacy, sound regulation and distributed capability matter as much as advancing the tech itself